

Unit 202 — Instructional Visual Review Package

Generated 2026-08-23T14:01:12.069Z

66 canonical instructional-visual variants, one per page, at a realistic mobile viewport width (390px).

This package is for REAL, human/external pixel-level review. No image in this package has been visually approved by this repository's tooling. Any "SIMULATED SEMANTIC" result referenced elsewhere in this project's reports comes from a deterministic mock pipeline that proves the QA architecture works end-to-end -- it is not a substitute for the review this package exists to support.

Each page shows: the actual rendered instructional SVG at mobile scale, its governed instructional purpose, which assessment-obligation/assertion/capability it is linked to, its TEACHING / ASSESSMENT / BOTH state, and the semantic contract's must-show / must-not-show requirements it is being checked against.

Show a single loop of wire rotating on a central axis between N and S poles, and that the loop's position relative to the field determines whether it is cutting flux at the maximum rate (near-peak EMF) or not at all (near-zero EMF).

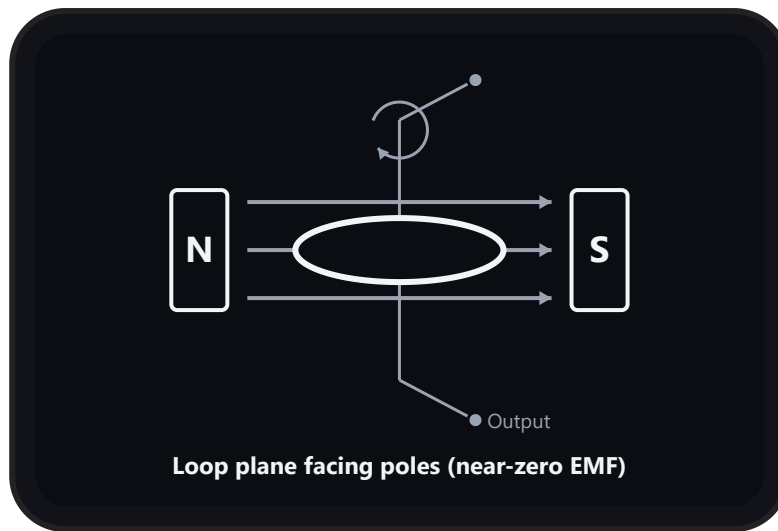


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

generator.rotating_loop (magnetic_field)

visual-contract.ac-generator-rotating-loop

visual-contract.ac-generator-rotating-loop@1::mode=both,rotation_phase=horizontal::both

electrical.emf_and_generation

EL-CONCEPT-AC-GENERATOR-001, EL-CONCEPT-SINE-WAVE-001

cap.emf.describe_ac_generation

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- a rotating loop between them
- a central rotation axis
- a minimal slip-ring/output connection

MUST NOT SHOW

- a waveform overlay (this pairs with the separate, existing graph.waveform_sine blueprint, never merged into one image)
- the static motor-principle diagram substituted for this rotating-loop concept (the defect this blueprint fixes)
- three-phase alternators, phasors or any vector-maths depiction

Show a single loop of wire rotating on a central axis between N and S poles, and that the loop's position relative to the field determines whether it is cutting flux at the maximum rate (near-peak EMF) or not at all (near-zero EMF).

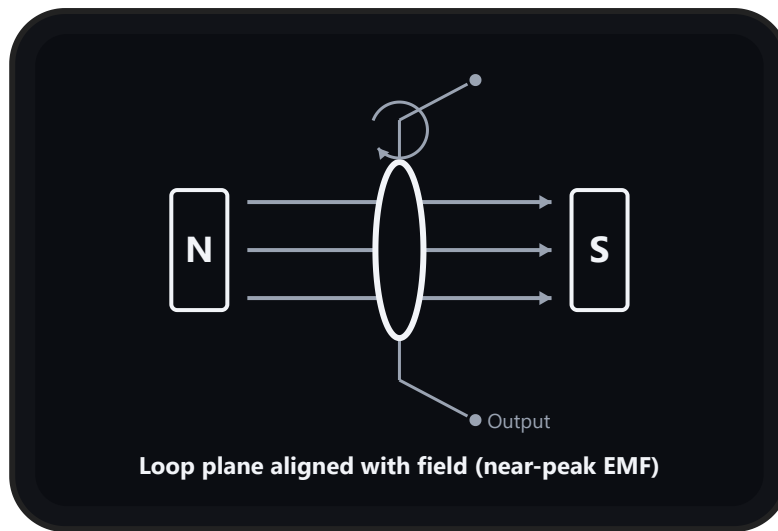


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

generator.rotating_loop (magnetic_field)

visual-contract.ac-generator-rotating-loop

visual-contract.ac-generator-rotating-loop@1::mode=both,rotation_phase=vertical::both

electrical.emf_and_generation

EL-CONCEPT-AC-GENERATOR-001, EL-CONCEPT-SINE-WAVE-001

cap.emf.describe_ac_generation

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- a rotating loop between them
- a central rotation axis
- a minimal slip-ring/output connection

MUST NOT SHOW

- a waveform overlay (this pairs with the separate, existing graph.waveform_sine blueprint, never merged into one image)
- the static motor-principle diagram substituted for this rotating-loop concept (the defect this blueprint fixes)
- three-phase alternators, phasors or any vector-maths depiction

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

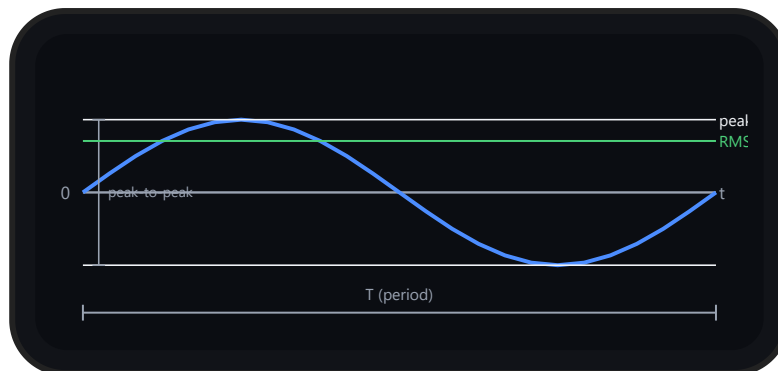


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=1,mode=both,show_peak_line=true,show_period_marker=true,show_rms_line=true::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

`EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001`

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from
- when `show_peak_line` is true: a labelled peak-to-peak bracket spanning from the peak line to the trough line

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- `EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001`)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

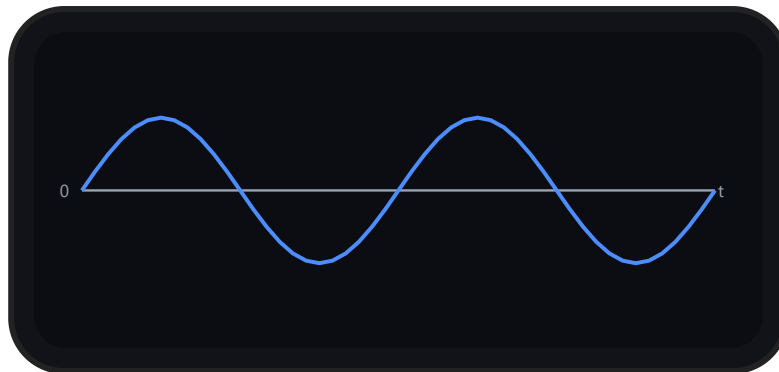


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=false,show_period_marker=false,show_rms_line=false::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from
- when `show_peak_line` is true: a labelled peak-to-peak bracket spanning from the peak line to the trough line

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

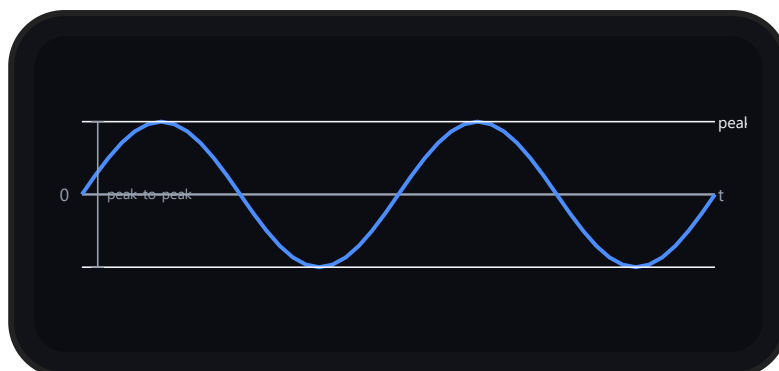


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=true,show_period_marker=false,show_rms_line=false::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from
- when `show_peak_line` is true: a labelled peak-to-peak bracket spanning from the peak line to the trough line

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

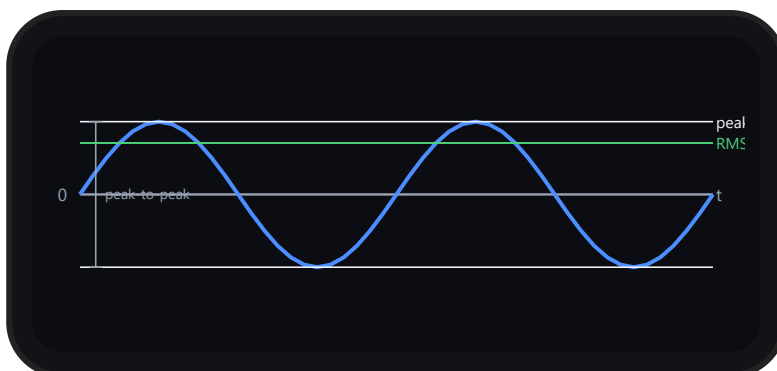


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=true,show_period_marker=false,show_rms_line=true::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

`EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001`

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from
- when `show_peak_line` is true: a labelled peak-to-peak bracket spanning from the peak line to the trough line

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- `EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001`)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

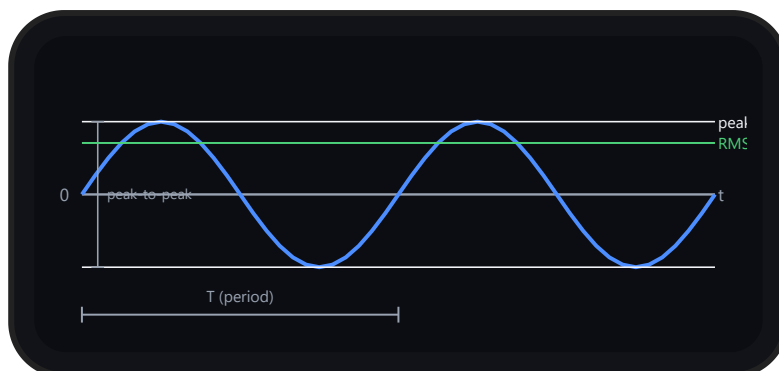


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=2,mode=both,show_peak_line=true,show_period_marker=true,show_rms_line=true::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

`EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001`

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from
- when `show_peak_line` is true: a labelled peak-to-peak bracket spanning from the peak line to the trough line

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- `EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001`)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show a sine AC waveform with correctly-positioned peak/RMS/period reference marks, teaching the $\text{RMS} \sim 0.707 \times \text{peak}$ relationship and the zero full-cycle average of symmetric AC.

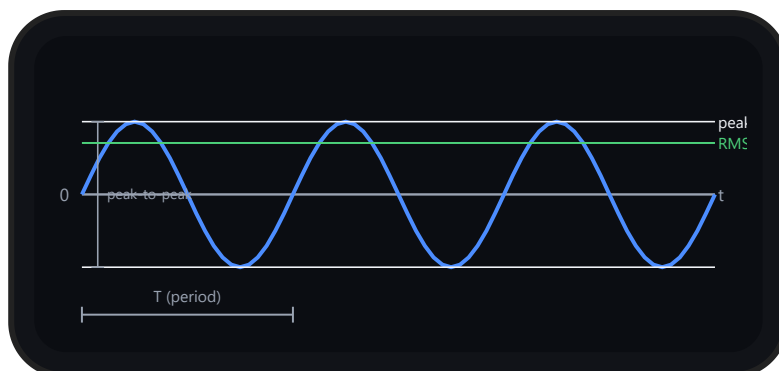


Diagram blueprint

`graph.waveform_sine (waveform)`

Visual semantic contract

`visual-contract.ac-waveform-sine`

Canonical variant id

`visual-contract.ac-waveform-sine@1::cycles_shown=3,mode=both,show_peak_line=true,show_period_marker=true,show_rms_line=true::both`

Linked assertion family

`electrical.ac_dc_waveforms`

Linked assertion identifiers

`EL-WAVEFORM-PERIODIC-TIME-001, EL-WAVEFORM-AMPLITUDE-001, EL-WAVEFORM-PEAK-TO-PEAK-001, EL-WAVEFORM-RMS-001, EL-WAVEFORM-AVERAGE-VALUE-001, EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001, EL-WAVEFORM-RMS-CALC-001, EL-WAVEFORM-RMS-PEAK-RELATIONSHIP-001`

Linked capability

`cap.waveform.identify_characteristic, cap.waveform.calculate_rms_peak`

Representation role

`essential`**MUST SHOW**

- the sine curve itself over `cycles_shown` full periods
- the zero/reference axis, which every other reference line is measured from
- when `show_peak_line` is true: a labelled peak-to-peak bracket spanning from the peak line to the trough line

MUST NOT SHOW

- an RMS line coincident with the peak line
- a non-zero horizontal 'average' reference line for a full symmetric cycle (the real signed average is zero -- `EL-WAVEFORM-AVERAGE-ZERO-INTERPRETATION-001`)
- any numeric peak/RMS/period value embedded in the artwork itself (values belong in question/lesson prompt text, per the symbolic-only convention every diagram in this folder follows)

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=capacitor,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint

electronics.component_symbol_card (component_symbol)

Visual semantic contract

visual-contract.electronic-component-symbol-card

Canonical variant id

visual-contract.electronic-component-symbol-card@1::component_type=diac,mode=both::both

Linked assertion family

electrical.electronic_components

Linked assertion identifiers

EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001

Linked capability

cap.electronic_components.recognise_principle,
cap.electronic_components.identify_application

Representation role

essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.

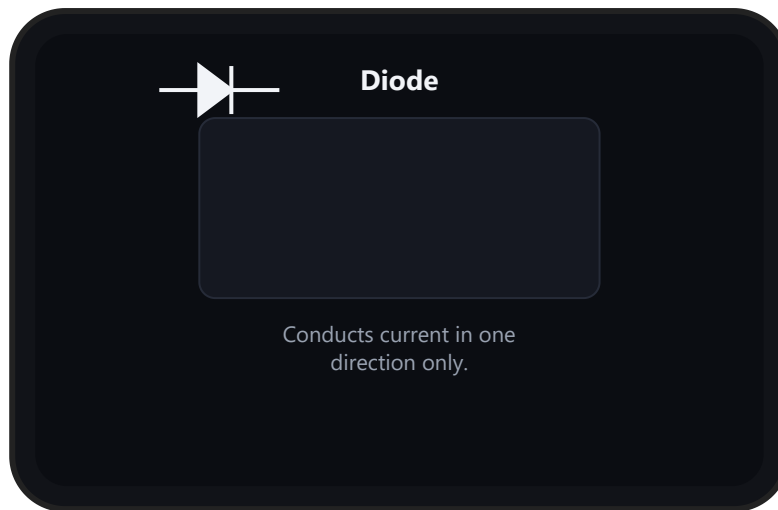


Diagram blueprint	<code>electronics.component_symbol_card (component_symbol)</code>
Visual semantic contract	<code>visual-contract.electronic-component-symbol-card</code>
Canonical variant id	<code>visual-contract.electronic-component-symbol-card@1::component_type=diode,mode=both::both</code>
Linked assertion family	<code>electrical.electronic_components</code>
Linked assertion identifiers	<code>EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001</code>
Linked capability	<code>cap.electronic_components.recognise_principle, cap.electronic_components.identify_application</code>
Representation role	<code>essential</code>

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=inverter,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=led,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint

electronics.component_symbol_card (component_symbol)

Visual semantic contract

visual-contract.electronic-component-symbol-card

Canonical variant id

visual-contract.electronic-component-symbol-card@1::component_type=photodiode,mode=both::both

Linked assertion family

electrical.electronic_components

Linked assertion identifiers

EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001

Linked capability

cap.electronic_components.recognise_principle,
cap.electronic_components.identify_application

Representation role

essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=rectifier,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=resistor,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=thermist or,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT- DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL- COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL- COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT- TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR- 001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=thyristor_scr,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=transistor,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.

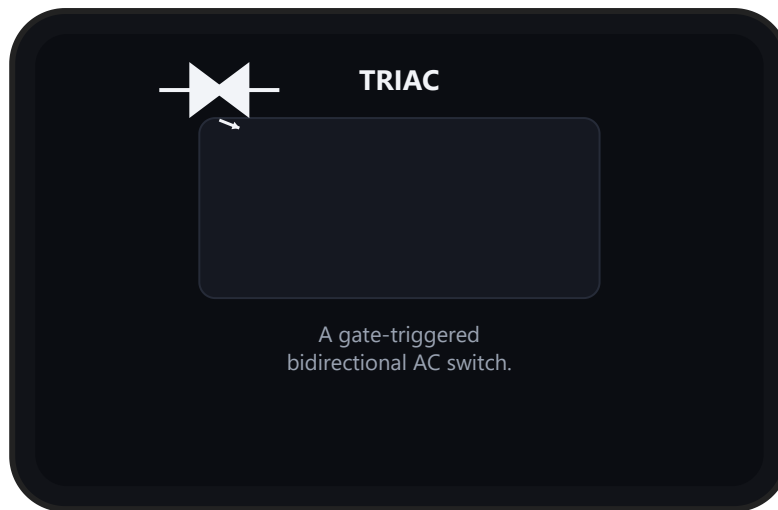


Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=triac,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show the governed BS EN 60617 / IEC 60617 schematic symbol (or, for rectifier/inverter, a functional-block illustration) for one named electronic component, for symbol recognition and at-a-glance function -- never detailed operating-principle prose.



Diagram blueprint	electronics.component_symbol_card (component_symbol)
Visual semantic contract	visual-contract.electronic-component-symbol-card
Canonical variant id	visual-contract.electronic-component-symbol-card@1::component_type=zener_diode,mode=both::both
Linked assertion family	electrical.electronic_components
Linked assertion identifiers	EL-COMPONENT-RESISTOR-001, EL-COMPONENT-CAPACITOR-001, EL-COMPONENT-DIODE-001, EL-COMPONENT-ZENER-DIODE-001, EL-COMPONENT-LED-001, EL-COMPONENT-PHOTODIODE-001, EL-COMPONENT-THERMISTOR-001, EL-COMPONENT-THERMISTOR-PTC-001, EL-COMPONENT-DIAC-001, EL-COMPONENT-TRIAC-001, EL-COMPONENT-TRANSISTOR-001, EL-COMPONENT-THYRISTOR-SCR-001, EL-COMPONENT-RECTIFIER-001, EL-COMPONENT-INVERTER-001
Linked capability	cap.electronic_components.recognise_principle, cap.electronic_components.identify_application
Representation role	essential

MUST SHOW

- the governed BS EN 60617 / IEC 60617 symbol (schematic symbol for 11 component_type values, functional block for rectifier/inverter) matching the requested component_type
- the component's name as a heading
- a short one-line functional caption

MUST NOT SHOW

- detailed operating-principle prose beyond the one-line caption (that stays in the lesson step's own text)
- any numeric component value
- the US/ANSI resistor zigzag or any other non-UK/IEC symbol convention

Show two meshed gears whose relative size represents the gear ratio, so the learner recognises the torque/speed trade-off a larger or smaller driven gear produces.

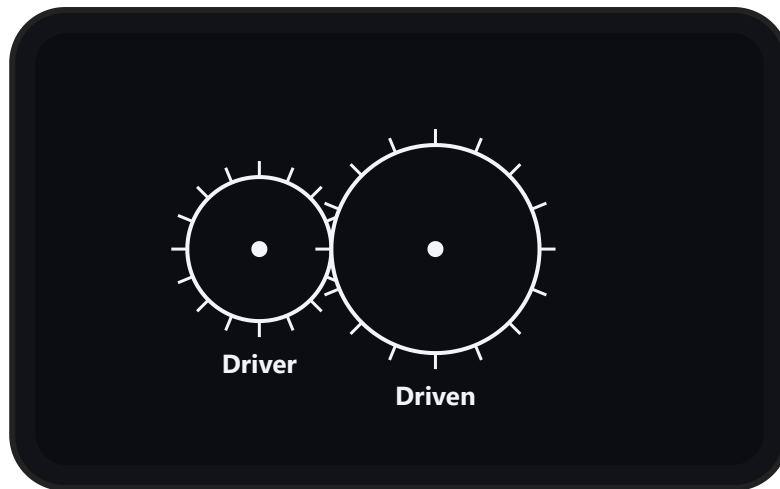


Diagram blueprint

`mechanical.gear_mesh (mechanical)`

Visual semantic contract

`visual-contract.gear-mesh-ratio`

Canonical variant id

`visual-contract.gear-mesh-ratio@1::mode=both,size_ratio=driven_larger::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-GEAR-001, FP-REL-GEAR-RATIO-001, FP-GEAR-SPEED-TORQUE-TRADEOFF-001`

Linked capability

`cap.foundational.gears.recognise`

Representation role

`essential`

MUST SHOW

- a driver gear, labelled
- a driven gear, labelled, tangent to (meshing with) the driver
- a ring of tooth marks around each gear's circumference

MUST NOT SHOW

- numeric radius or tooth-count values (recognition-depth only -- no gear-ratio calculation blueprint exists)
- rotation-direction arrows as a primary element (not the governed teaching point)

Show two meshed gears whose relative size represents the gear ratio, so the learner recognises the torque/speed trade-off a larger or smaller driven gear produces.

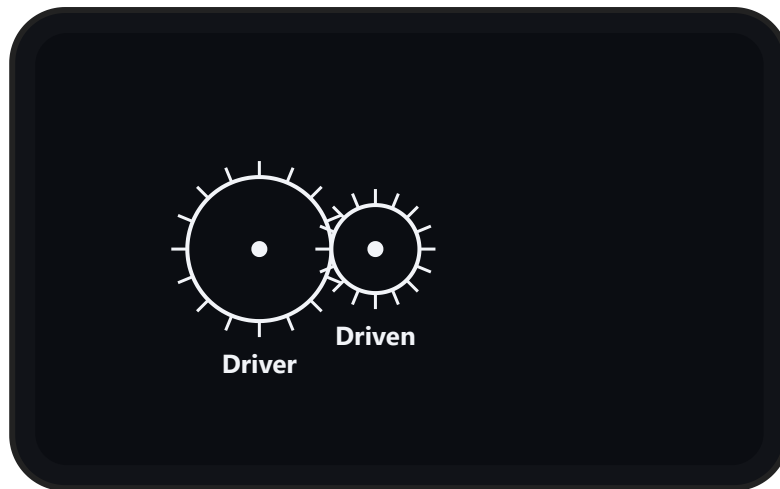


Diagram blueprint	<code>mechanical.gear_mesh (mechanical)</code>
Visual semantic contract	<code>visual-contract.gear-mesh-ratio</code>
Canonical variant id	<code>visual-contract.gear-mesh-ratio@1::mode=both,size_ratio=driven_smaller::both</code>
Linked assertion family	<code>foundational.levers_mechanical_advantage</code>
Linked assertion identifiers	<code>FP-CONCEPT-GEAR-001, FP-REL-GEAR-RATIO-001, FP-GEAR-SPEED-TORQUE-TRADEOFF-001</code>
Linked capability	<code>cap.foundational.gears.recognise</code>
Representation role	<code>essential</code>

MUST SHOW

- a driver gear, labelled
- a driven gear, labelled, tangent to (meshing with) the driver
- a ring of tooth marks around each gear's circumference

MUST NOT SHOW

- numeric radius or tooth-count values (recognition-depth only -- no gear-ratio calculation blueprint exists)
- rotation-direction arrows as a primary element (not the governed teaching point)

Show two meshed gears whose relative size represents the gear ratio, so the learner recognises the torque/speed trade-off a larger or smaller driven gear produces.

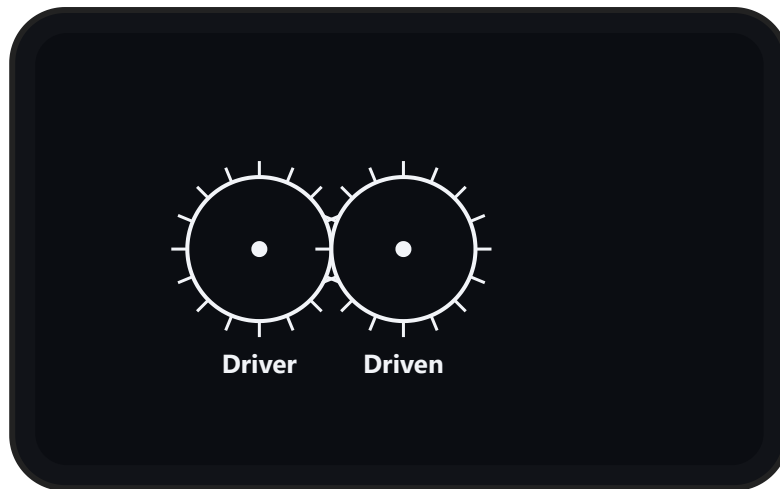


Diagram blueprint

`mechanical.gear_mesh (mechanical)`

Visual semantic contract

`visual-contract.gear-mesh-ratio`

Canonical variant id

`visual-contract.gear-mesh-ratio@1::mode=both,size_ratio=equal::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-GEAR-001, FP-REL-GEAR-RATIO-001, FP-GEAR-SPEED-TORQUE-TRADEOFF-001`

Linked capability

`cap.foundational.gears.recognise`

Representation role

`essential`

MUST SHOW

- a driver gear, labelled
- a driven gear, labelled, tangent to (meshing with) the driver
- a ring of tooth marks around each gear's circumference

MUST NOT SHOW

- numeric radius or tooth-count values (recognition-depth only -- no gear-ratio calculation blueprint exists)
- rotation-direction arrows as a primary element (not the governed teaching point)

Teach correct voltmeter-parallel / ammeter-series connection (including, in TEACHING mode, deliberate non-standard comparison examples), and that ohmmeter use requires a de-energised circuit, with isolation of the individual component from other parallel paths where needed for an accurate reading.

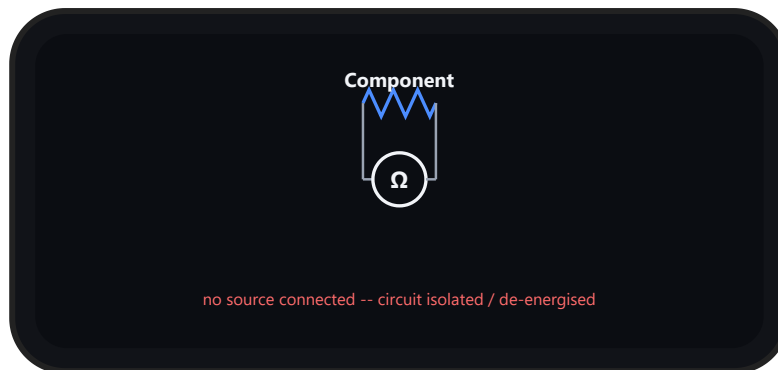


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=isolated,instrument_type=ohmmeter,mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001, EL-INSTRUMENT-OHMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested `connection_style`
- the component/path being measured
- in TEACHING or BOTH mode: an explicit on-diagram caption stating whether the shown combination matches the standard, correct connection method for that instrument (never silently implying a miswiring is standard practice)

MUST NOT SHOW

- a voltmeter spliced into the current path, or an ammeter drawn as a side branch, in TEACHING or BOTH mode, without the caption identifying whether it is the standard connection
- the standard/non-standard caption in ASSESSMENT mode (it directly states `instrumentation.recognise_connection`'s own answer -- series or parallel)
- for instrument_type 'ohmmeter', any energised-source symbol in a closed loop with the meter -- ohmmeter use requires an isolated, de-energised component

Teach correct voltmeter-parallel / ammeter-series connection (including, in TEACHING mode, deliberate non-standard comparison examples), and that ohmmeter use requires a de-energised circuit, with isolation of the individual component from other parallel paths where needed for an accurate reading.

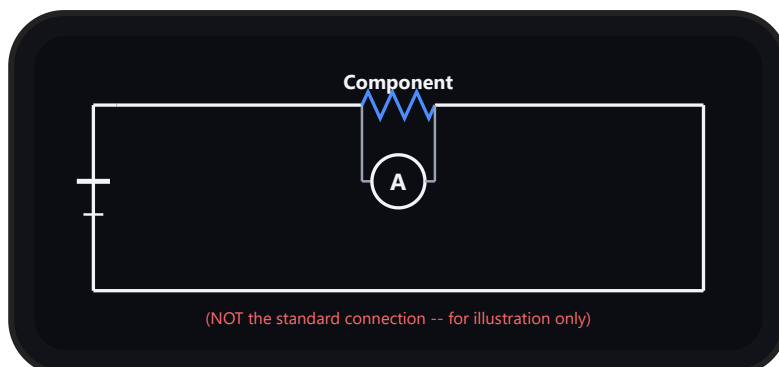


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=parallel, instrument_type=ammeter, mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001, EL-INSTRUMENT-OHMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested `connection_style`
- the component/path being measured
- in TEACHING or BOTH mode: an explicit on-diagram caption stating whether the shown combination matches the standard, correct connection method for that instrument (never silently implying a miswiring is standard practice)

MUST NOT SHOW

- a voltmeter spliced into the current path, or an ammeter drawn as a side branch, in TEACHING or BOTH mode, without the caption identifying whether it is the standard connection
- the standard/non-standard caption in ASSESSMENT mode (it directly states `instrumentation.recognise_connection`'s own answer -- series or parallel)
- for instrument_type 'ohmmeter', any energised-source symbol in a closed loop with the meter -- ohmmeter use requires an isolated, de-energised component

Teach correct voltmeter-parallel / ammeter-series connection (including, in TEACHING mode, deliberate non-standard comparison examples), and that ohmmeter use requires a de-energised circuit, with isolation of the individual component from other parallel paths where needed for an accurate reading.

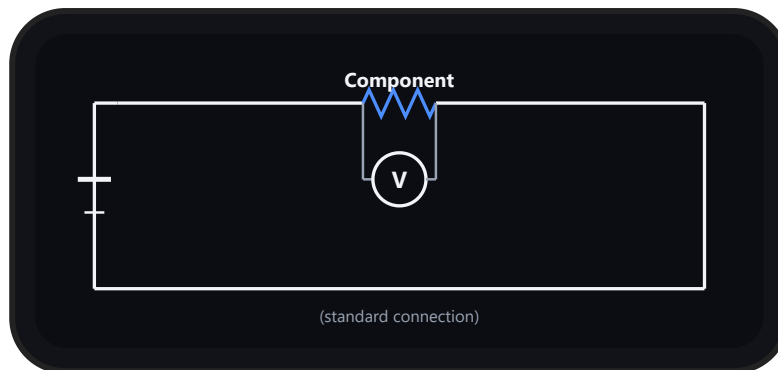


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=parallel, instrument_type=voltmeter, mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001, EL-INSTRUMENT-OHMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested `connection_style`
- the component/path being measured
- in TEACHING or BOTH mode: an explicit on-diagram caption stating whether the shown combination matches the standard, correct connection method for that instrument (never silently implying a miswiring is standard practice)

MUST NOT SHOW

- a voltmeter spliced into the current path, or an ammeter drawn as a side branch, in TEACHING or BOTH mode, without the caption identifying whether it is the standard connection
- the standard/non-standard caption in ASSESSMENT mode (it directly states `instrumentation.recognise_connection`'s own answer -- series or parallel)
- for instrument_type 'ohmmeter', any energised-source symbol in a closed loop with the meter -- ohmmeter use requires an isolated, de-energised component

Teach correct voltmeter-parallel / ammeter-series connection (including, in TEACHING mode, deliberate non-standard comparison examples), and that ohmmeter use requires a de-energised circuit, with isolation of the individual component from other parallel paths where needed for an accurate reading.

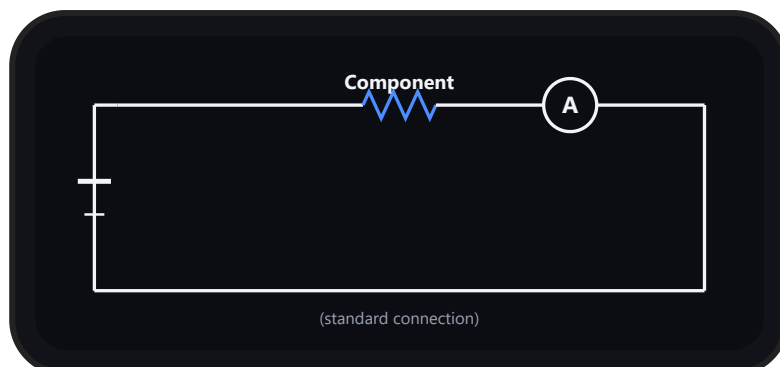


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=series,instrument_type=ammeter,mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001, EL-INSTRUMENT-OHMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested `connection_style`
- the component/path being measured
- in TEACHING or BOTH mode: an explicit on-diagram caption stating whether the shown combination matches the standard, correct connection method for that instrument (never silently implying a miswiring is standard practice)

MUST NOT SHOW

- a voltmeter spliced into the current path, or an ammeter drawn as a side branch, in TEACHING or BOTH mode, without the caption identifying whether it is the standard connection
- the standard/non-standard caption in ASSESSMENT mode (it directly states `instrumentation.recognise_connection`'s own answer -- series or parallel)
- for instrument_type 'ohmmeter', any energised-source symbol in a closed loop with the meter -- ohmmeter use requires an isolated, de-energised component

Teach correct voltmeter-parallel / ammeter-series connection (including, in TEACHING mode, deliberate non-standard comparison examples), and that ohmmeter use requires a de-energised circuit, with isolation of the individual component from other parallel paths where needed for an accurate reading.

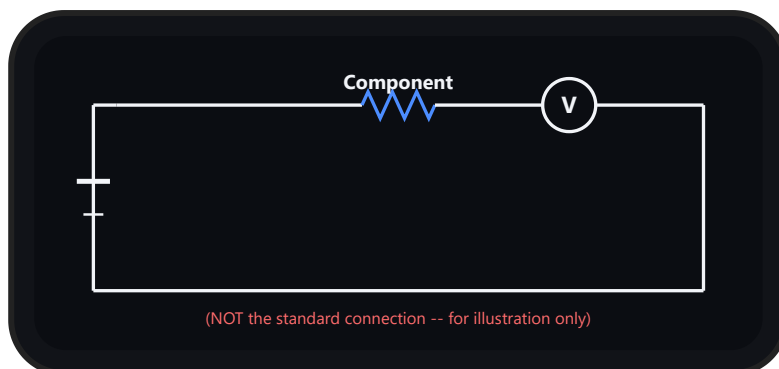


Diagram blueprint	<code>instrument.measurement_connection (instrument_connection)</code>
Visual semantic contract	<code>visual-contract.instrument-measurement-connection</code>
Canonical variant id	<code>visual-contract.instrument-measurement-connection@1::connection_style=series,instrument_type=voltmeter,mode=both::both</code>
Linked assertion family	<code>electrical.instrumentation</code>
Linked assertion identifiers	<code>EL-INSTRUMENT-VOLTMETER-001, EL-INSTRUMENT-AMMETER-001, EL-INSTRUMENT-OHMMETER-001</code>
Linked capability	<code>cap.instrumentation.recognise_connection</code>
Representation role	<code>essential</code>

MUST SHOW

- the instrument symbol (circle containing V/A/Omega) at the correct topological position for the requested `connection_style`
- the component/path being measured
- in TEACHING or BOTH mode: an explicit on-diagram caption stating whether the shown combination matches the standard, correct connection method for that instrument (never silently implying a miswiring is standard practice)

MUST NOT SHOW

- a voltmeter spliced into the current path, or an ammeter drawn as a side branch, in TEACHING or BOTH mode, without the caption identifying whether it is the standard connection
- the standard/non-standard caption in ASSESSMENT mode (it directly states `instrumentation.recognise_connection`'s own answer -- series or parallel)
- for instrument_type 'ohmmeter', any energised-source symbol in a closed loop with the meter -- ohmmeter use requires an isolated, de-energised component

Show a lever's pivot, effort and load positioned along a bar so the learner can identify the lever class (I/II/III) from the arrangement itself, and (optionally) the effort-arm/load-arm distances the moment-balance relationship depends on.

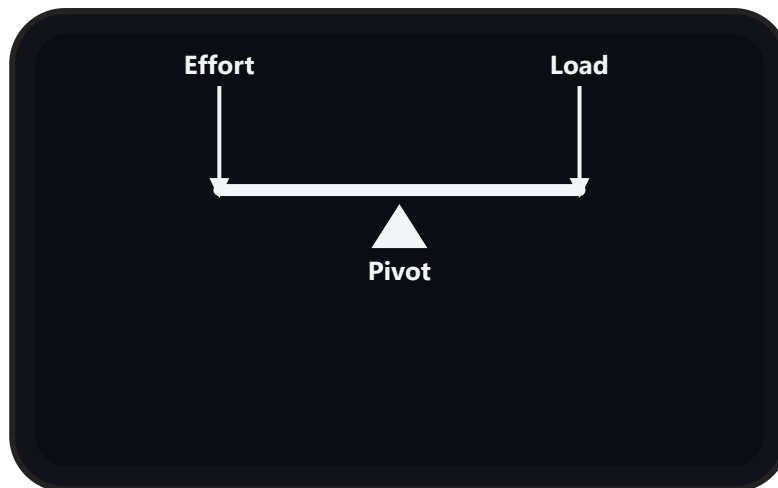


Diagram blueprint

`mechanical.lever_arrangement (mechanical)`

Visual semantic contract

`visual-contract.lever-class-arrangement`

Canonical variant id

`visual-contract.lever-class-arrangement@1::lever_class=class_1,mode=both,show_distances=false::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-LEVER-PRINCIPLE-001, FP-LEVER-CLASS-I-001, FP-LEVER-CLASS-II-001, FP-LEVER-CLASS-III-001, FP-REL-LEVER-BALANCE-001`

Linked capability

`cap.foundational.levers.recognise, cap.foundational.levers.calculate`

Representation role

`essential`**MUST SHOW**

- a single horizontal bar
- the pivot (triangle fulcrum symbol) at its class-determined position
- an effort arrow and a load arrow, each labelled in text, at their class-determined positions

MUST NOT SHOW

- numeric force or distance values (symbolic d_e/d_l labels only, per CC-05A valueEmbedding policy)

Show a lever's pivot, effort and load positioned along a bar so the learner can identify the lever class (I/II/III) from the arrangement itself, and (optionally) the effort-arm/load-arm distances the moment-balance relationship depends on.

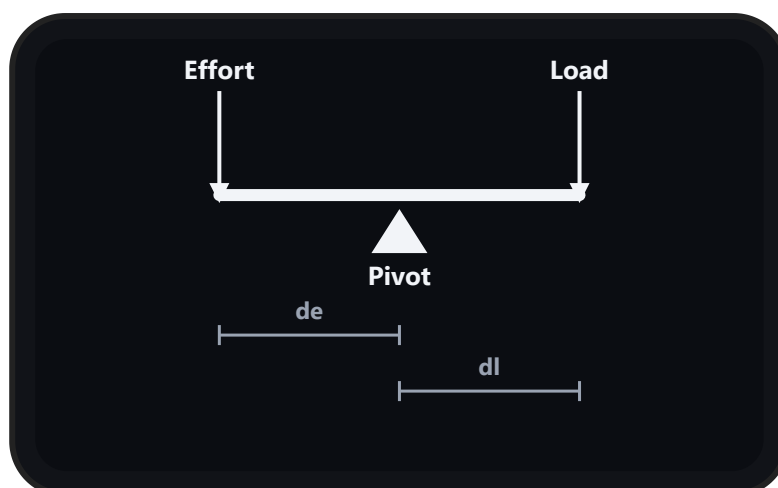


Diagram blueprint

`mechanical.lever_arrangement (mechanical)`

Visual semantic contract

`visual-contract.lever-class-arrangement`

Canonical variant id

`visual-contract.lever-class-arrangement@1::lever_class=class_1,mode=both,show_distances=true::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-LEVER-PRINCIPLE-001, FP-LEVER-CLASS-I-001, FP-LEVER-CLASS-II-001, FP-LEVER-CLASS-III-001, FP-REL-LEVER-BALANCE-001`

Linked capability

`cap.foundational.levers.recognise, cap.foundational.levers.calculate`

Representation role

`essential`**MUST SHOW**

- a single horizontal bar
- the pivot (triangle fulcrum symbol) at its class-determined position
- an effort arrow and a load arrow, each labelled in text, at their class-determined positions

MUST NOT SHOW

- numeric force or distance values (symbolic d_e/d_l labels only, per CC-05A valueEmbedding policy)

Show a lever's pivot, effort and load positioned along a bar so the learner can identify the lever class (I/II/III) from the arrangement itself, and (optionally) the effort-arm/load-arm distances the moment-balance relationship depends on.

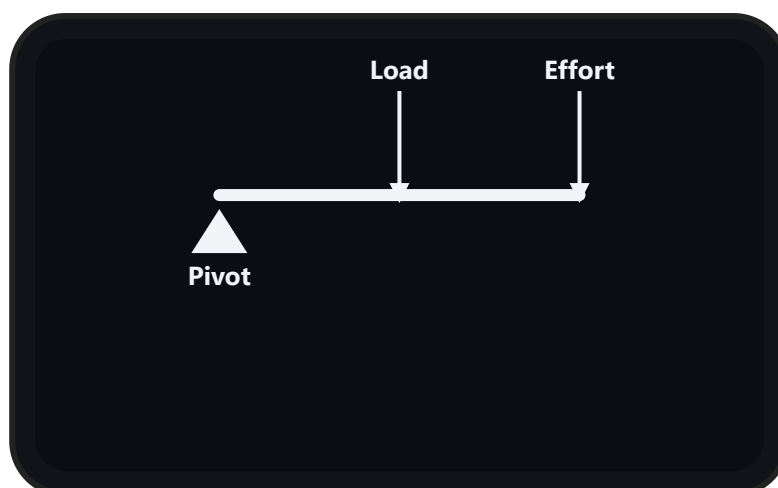


Diagram blueprint

`mechanical.lever_arrangement (mechanical)`

Visual semantic contract

`visual-contract.lever-class-arrangement`

Canonical variant id

`visual-contract.lever-class-arrangement@1::lever_class=class_2,mode=both,show_distances=false::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-LEVER-PRINCIPLE-001, FP-LEVER-CLASS-I-001, FP-LEVER-CLASS-II-001, FP-LEVER-CLASS-III-001, FP-REL-LEVER-BALANCE-001`

Linked capability

`cap.foundational.levers.recognise, cap.foundational.levers.calculate`

Representation role

`essential`**MUST SHOW**

- a single horizontal bar
- the pivot (triangle fulcrum symbol) at its class-determined position
- an effort arrow and a load arrow, each labelled in text, at their class-determined positions

MUST NOT SHOW

- numeric force or distance values (symbolic d_e/d_l labels only, per CC-05A valueEmbedding policy)

Show a lever's pivot, effort and load positioned along a bar so the learner can identify the lever class (I/II/III) from the arrangement itself, and (optionally) the effort-arm/load-arm distances the moment-balance relationship depends on.

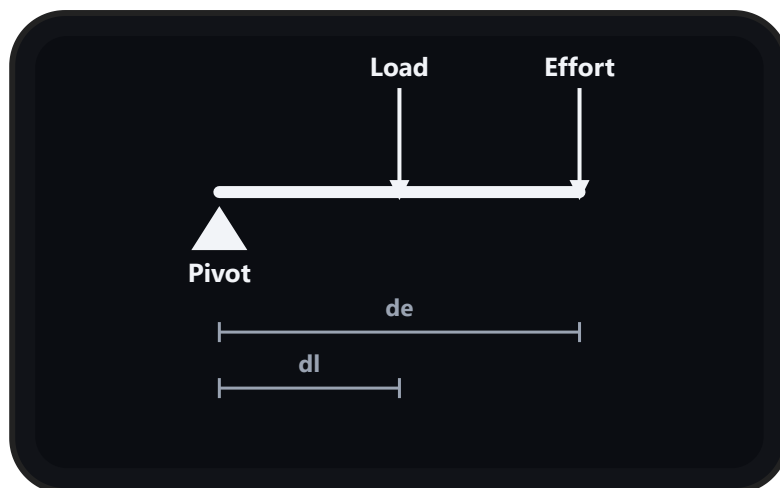


Diagram blueprint

`mechanical.lever_arrangement (mechanical)`

Visual semantic contract

`visual-contract.lever-class-arrangement`

Canonical variant id

`visual-contract.lever-class-arrangement@1::lever_class=class_2,mode=both,show_distances=true::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-LEVER-PRINCIPLE-001, FP-LEVER-CLASS-I-001, FP-LEVER-CLASS-II-001, FP-LEVER-CLASS-III-001, FP-REL-LEVER-BALANCE-001`

Linked capability

`cap.foundational.levers.recognise, cap.foundational.levers.calculate`

Representation role

`essential`**MUST SHOW**

- a single horizontal bar
- the pivot (triangle fulcrum symbol) at its class-determined position
- an effort arrow and a load arrow, each labelled in text, at their class-determined positions

MUST NOT SHOW

- numeric force or distance values (symbolic d_e/d_l labels only, per CC-05A valueEmbedding policy)

Show a lever's pivot, effort and load positioned along a bar so the learner can identify the lever class (I/II/III) from the arrangement itself, and (optionally) the effort-arm/load-arm distances the moment-balance relationship depends on.

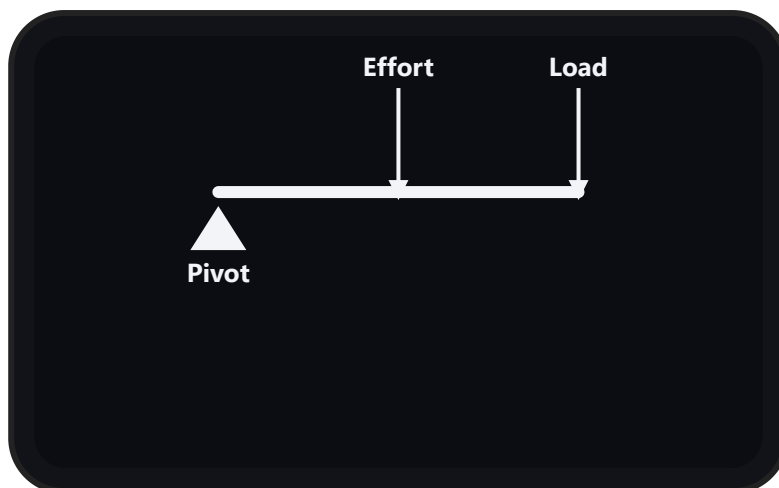


Diagram blueprint

mechanical.lever_arrangement (mechanical)

Visual semantic contract

visual-contract.lever-class-arrangement

Canonical variant id

visual-contract.lever-class-arrangement@1::lever_class=class_3,mode=both,show_distances=false::both

Linked assertion family

foundational.levers_mechanical_advantage

Linked assertion identifiers

FP-CONCEPT-LEVER-PRINCIPLE-001, FP-LEVER-CLASS-I-001, FP-LEVER-CLASS-II-001, FP-LEVER-CLASS-III-001, FP-REL-LEVER-BALANCE-001

Linked capability

cap.foundational.levers.recognise, cap.foundational.levers.calculate

Representation role

essential

MUST SHOW

- a single horizontal bar
- the pivot (triangle fulcrum symbol) at its class-determined position
- an effort arrow and a load arrow, each labelled in text, at their class-determined positions

MUST NOT SHOW

- numeric force or distance values (symbolic de/dl labels only, per CC-05A valueEmbedding policy)

Show a lever's pivot, effort and load positioned along a bar so the learner can identify the lever class (I/II/III) from the arrangement itself, and (optionally) the effort-arm/load-arm distances the moment-balance relationship depends on.

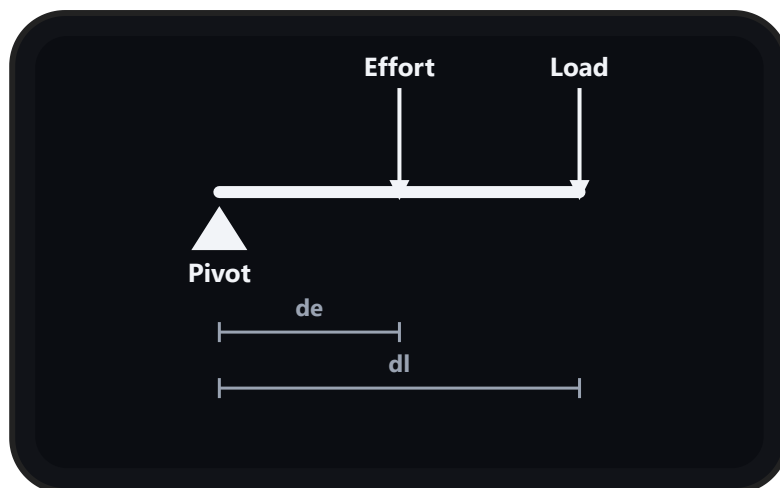


Diagram blueprint

`mechanical.lever_arrangement (mechanical)`

Visual semantic contract

`visual-contract.lever-class-arrangement`

Canonical variant id

`visual-contract.lever-class-arrangement@1::lever_class=class_3,mode=both,show_distances=true::both`

Linked assertion family

`foundational.levers_mechanical_advantage`

Linked assertion identifiers

`FP-CONCEPT-LEVER-PRINCIPLE-001, FP-LEVER-CLASS-I-001, FP-LEVER-CLASS-II-001, FP-LEVER-CLASS-III-001, FP-REL-LEVER-BALANCE-001`

Linked capability

`cap.foundational.levers.recognise, cap.foundational.levers.calculate`

Representation role

`essential`**MUST SHOW**

- a single horizontal bar
- the pivot (triangle fulcrum symbol) at its class-determined position
- an effort arrow and a load arrow, each labelled in text, at their class-determined positions

MUST NOT SHOW

- numeric force or distance values (symbolic d_e/d_l labels only, per CC-05A valueEmbedding policy)

Show magnetic flux as the field-line pattern from N to S, and (when density_comparison is shown) flux density as the same flux concentrated over different cross-sectional areas.

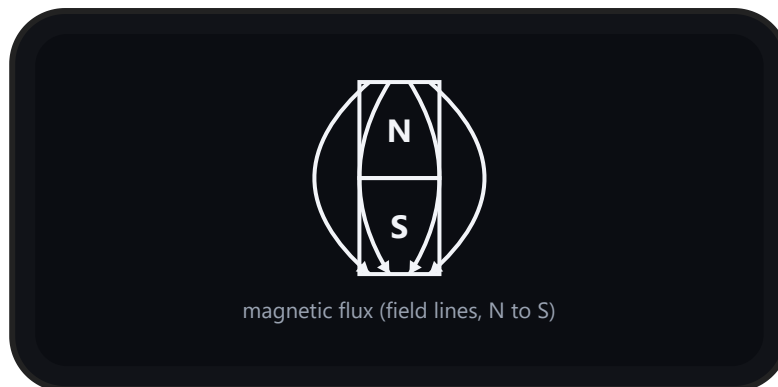


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.flux_field_lines (magnetic_field)

visual-contract.magnetic-flux-density-comparison

visual-contract.magnetic-flux-density-comparison@1::density_comparison=false,mode=both::both

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FLUX-001, EL-CONCEPT-MAGNETIC-FLUX-DENSITY-001

cap.magnetism.recognise_concept

supporting

MUST SHOW

- a labelled north pole
- a labelled south pole
- curved field lines from north to south

MUST NOT SHOW

- any numeric flux or flux-density value or formula (purely conceptual, per CC-05A valueEmbedding policy)

Show magnetic flux as the field-line pattern from N to S, and (when density_comparison is shown) flux density as the same flux concentrated over different cross-sectional areas.

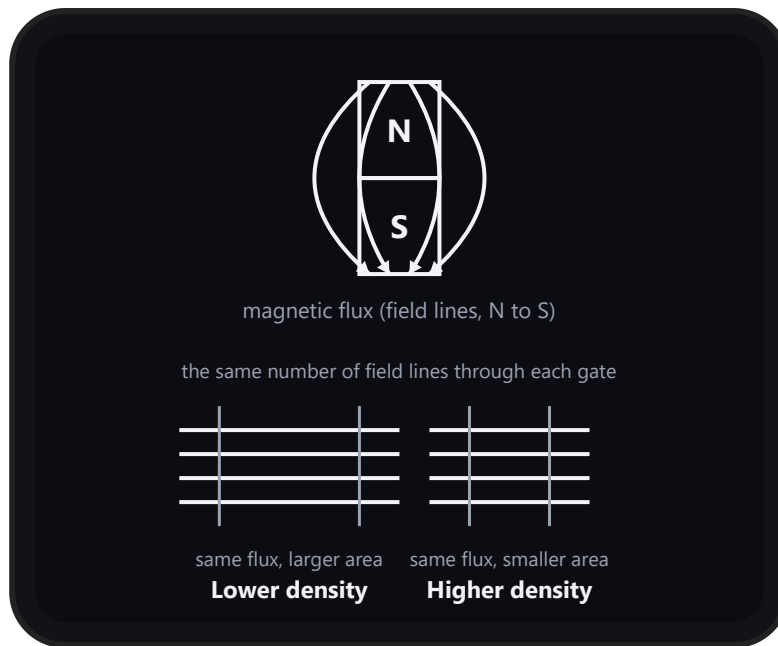


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.flux_field_lines (magnetic_field)

visual-contract.magnetic-flux-density-comparison

visual-contract.magnetic-flux-density-comparison@1::density_comparison=true,mode=both::both

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FLUX-001, EL-CONCEPT-MAGNETIC-FLUX-DENSITY-001

cap.magnetism.recognise_concept

supporting

MUST SHOW

- a labelled north pole
- a labelled south pole
- curved field lines from north to south

MUST NOT SHOW

- any numeric flux or flux-density value or formula (purely conceptual, per CC-05A valueEmbedding policy)

Teach that like magnetic poles repel and unlike poles attract, from the pole labels on two facing bar magnets.



Diagram blueprint

magnetic.pole_interaction (magnetic_field)

Visual semantic contract

visual-contract.magnetic-pole-interaction

Canonical variant id

visual-contract.magnetic-pole-interaction@1::mode=assessment,pole_pairing=like_poles_facing::assessment

Linked assertion family

electrical.magnetism_and_electromagnetism

Linked assertion identifiers

EL-CONCEPT-MAGNETISM-001

Linked capability

cap.magnetism.recognise_attraction_repulsion

Representation role

essential

MUST SHOW

- both poles of the left magnet, labelled
- both poles of the right magnet, labelled
- a visible central gap between the two facing poles

MUST NOT SHOW

- the force-arrows/attract-repel label in assessment mode (it is the assessed answer)

Teach that like magnetic poles repel and unlike poles attract, from the pole labels on two facing bar magnets.

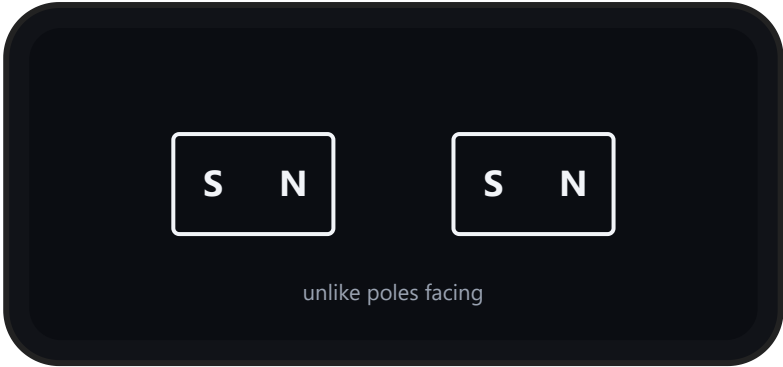


Diagram blueprint	magnetic.pole_interaction (magnetic_field)
Visual semantic contract	visual-contract.magnetic-pole-interaction
Canonical variant id	visual-contract.magnetic-pole-interaction@1::mode=assessment,pole_pairing=unlike_poles_facing::assessment
Linked assertion family	electrical.magnetism_and_electromagnetism
Linked assertion identifiers	EL-CONCEPT-MAGNETISM-001
Linked capability	cap.magnetism.recognise_attraction_repulsion
Representation role	essential

MUST SHOW

- both poles of the left magnet, labelled
- both poles of the right magnet, labelled
- a visible central gap between the two facing poles

MUST NOT SHOW

- the force-arrows/attract-repel label in assessment mode (it is the assessed answer)

Teach that like magnetic poles repel and unlike poles attract, from the pole labels on two facing bar magnets.

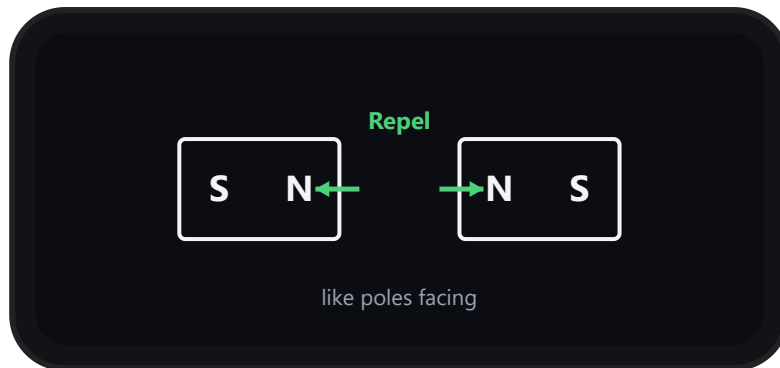


Diagram blueprint

magnetic.pole_interaction (magnetic_field)

Visual semantic contract

visual-contract.magnetic-pole-interaction

Canonical variant id

visual-contract.magnetic-pole-interaction@1::mode=teaching,pole_pairing=like_poles_facing,show_pole_force=true::teaching

Linked assertion family

electrical.magnetism_and_electromagnetism

Linked assertion identifiers

EL-CONCEPT-MAGNETISM-001

Linked capability

cap.magnetism.recognise_attraction_repulsion

Representation role

essential

MUST SHOW

- both poles of the left magnet, labelled
- both poles of the right magnet, labelled
- a visible central gap between the two facing poles

MUST NOT SHOW

- the force-arrows/attract-repel label in assessment mode (it is the assessed answer)

Teach that like magnetic poles repel and unlike poles attract, from the pole labels on two facing bar magnets.

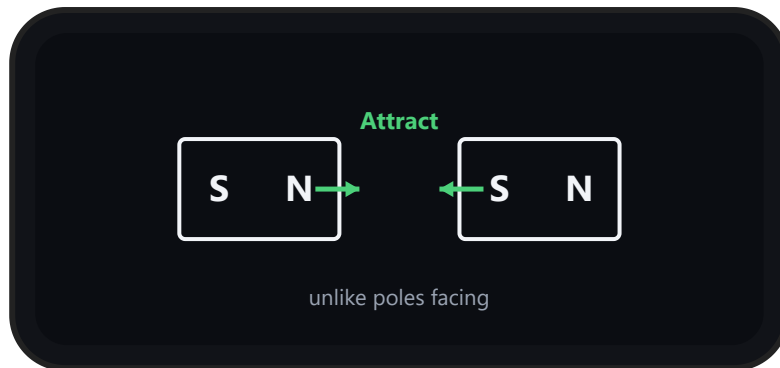


Diagram blueprint

magnetic.pole_interaction (magnetic_field)

Visual semantic contract

visual-contract.magnetic-pole-interaction

Canonical variant id

visual-contract.magnetic-pole-interaction@1::mode=teaching,pole_pairing=unlike_poles_facing,show_pole_force=true::teaching

Linked assertion family

electrical.magnetism_and_electromagnetism

Linked assertion identifiers

EL-CONCEPT-MAGNETISM-001

Linked capability

cap.magnetism.recognise_attraction_repulsion

Representation role

essential

MUST SHOW

- both poles of the left magnet, labelled
- both poles of the right magnet, labelled
- a visible central gap between the two facing poles

MUST NOT SHOW

- the force-arrows/attract-repel label in assessment mode (it is the assessed answer)

Show that the conductor length, its velocity and the magnetic field are mutually perpendicular -- the given geometric fact behind the simple $\mathcal{E} = B l v$ formula.

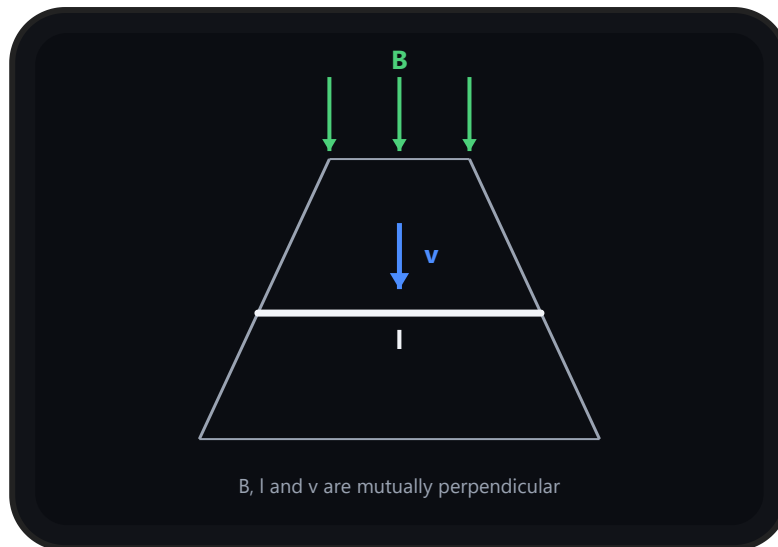


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

emf.motional_emf_geometry (magnetic_field)

visual-contract.motional-emf-geometry

visual-contract.motional-emf-geometry@1::mode=both::both

electrical.emf_and_generation

EL-REL-INDUCED-EMF-001

cap.emf.calculate_motional_emf

essential

MUST SHOW

- two parallel rails suggesting depth
- a conductor rod lying across the rails, labelled l
- a velocity arrow along the rails, labelled v
- field arrows perpendicular to the rail-plane, labelled B

MUST NOT SHOW

- any oblique/non-perpendicular arrangement
- a numeric EMF value or the formula itself

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

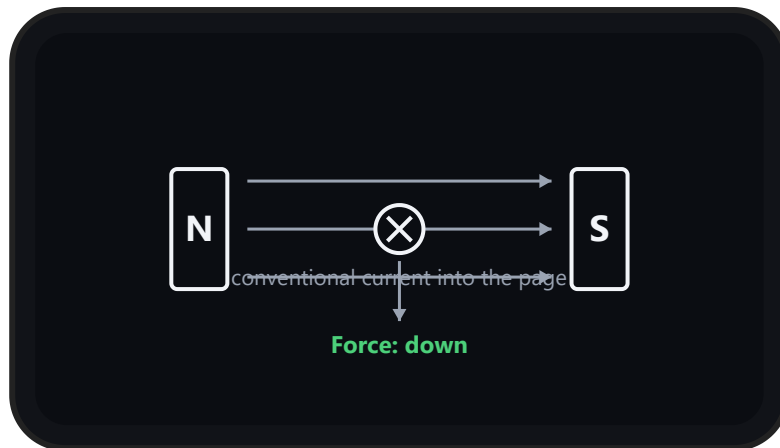


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,force_direction=down,mode=teaching,pole_labels=N_S_horizontal,show_force_arrow=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

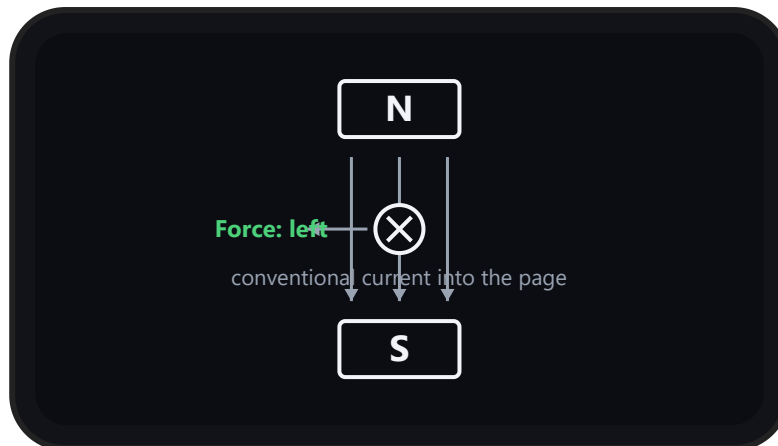


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,force_direction=left,mode=teaching,pole_labels=N_S_vertical,show_force_arrow=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

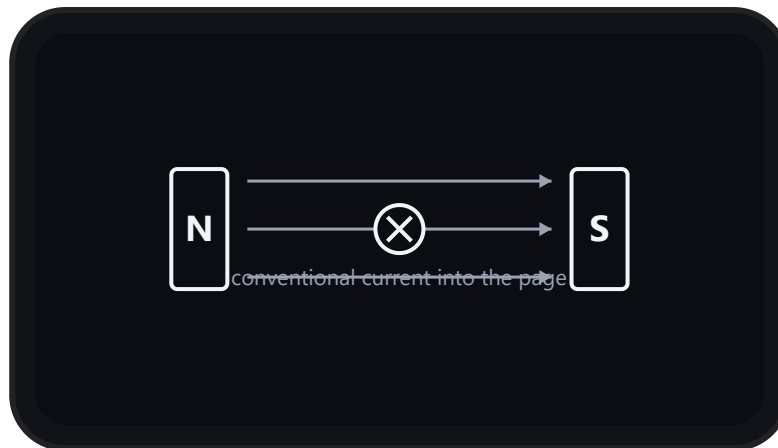


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,mode=assessment,pole_labels=N_S_horizontal,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

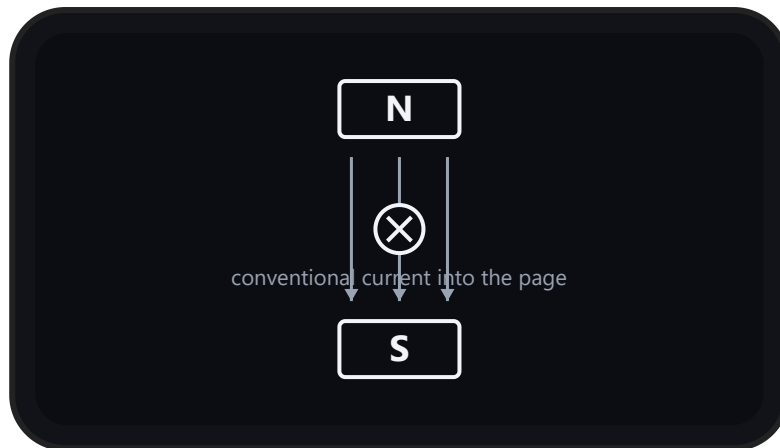


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=into_page,mode=assessment,pole_labels=N_S_vertical,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

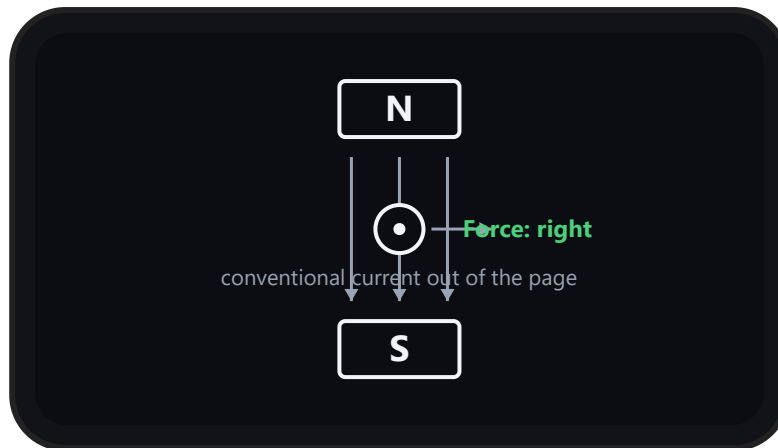


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

```
visual-contract.motor-principle-force@1::current_direction=out_of_page,force_direction=right,mode=teaching,pole_labels=N_S_vertical,show_force_arrow=true::teaching
```

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

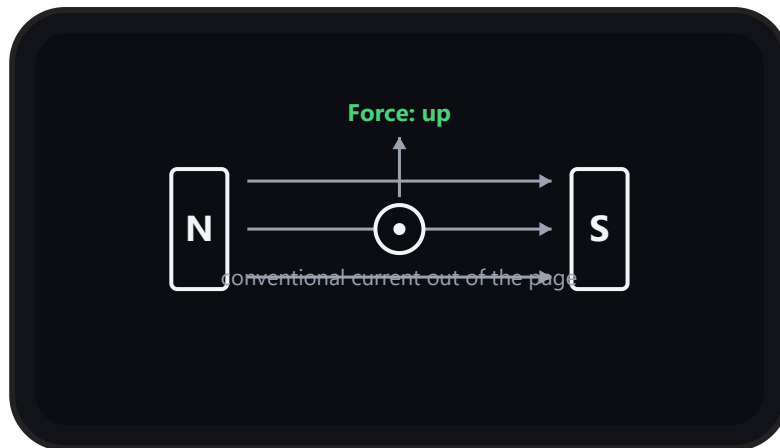


Diagram blueprint

Visual semantic contract

Canonical variant id

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,force_direction=up,mode=teaching,pole_labels=N_S_horizontal,show_force_arrow=true::teaching

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

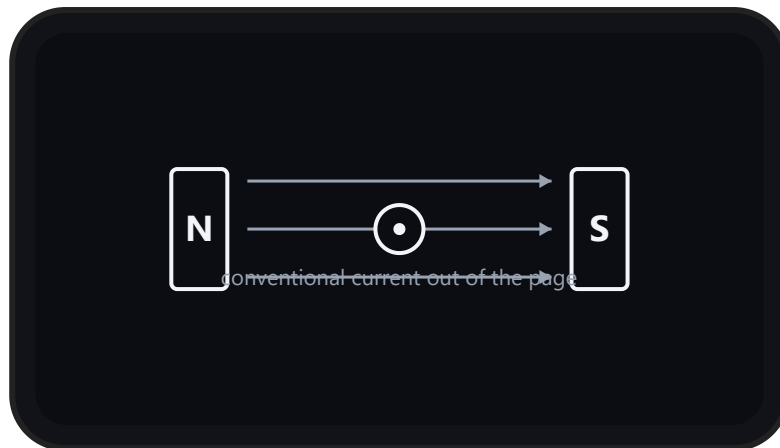


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,mode=assessment,pole_labels=N_S_horizontal,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Teach the motor principle: the direction of the force on a current-carrying conductor in a magnetic field.

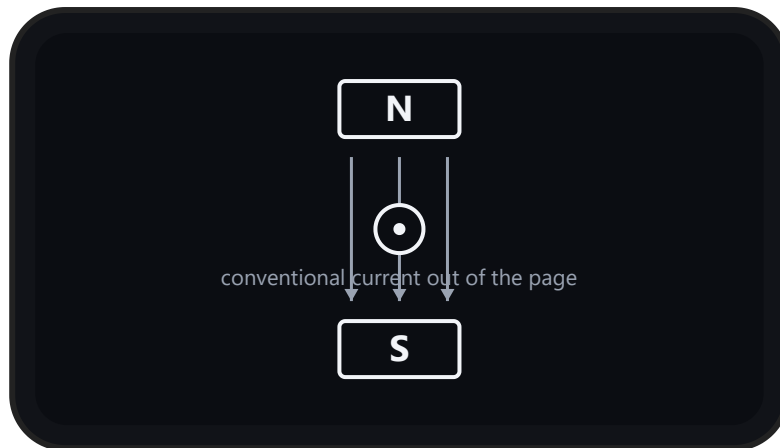


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

motor.force_field_current (magnetic_field)

visual-contract.motor-principle-force

visual-contract.motor-principle-force@1::current_direction=out_of_page,mode=assessment,pole_labels=N_S_vertical,show_force_arrow=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-FORCE-ON-CONDUCTOR-001, EL-CONCEPT-MOTOR-PRINCIPLE-001

cap.magnetism.interpret_force_direction

essential

MUST SHOW

- a labelled north pole
- a labelled south pole
- field-direction arrows from north to south
- the conductor cross-section between the poles

MUST NOT SHOW

- a hand/grip-rule visual (this is the motor-principle diagram, distinct from the right-hand-grip-rule diagram -- the two concepts must never be conflated in one image)
- the force arrow/label in assessment mode

Show resistors connected as independent branches between two shared rails.

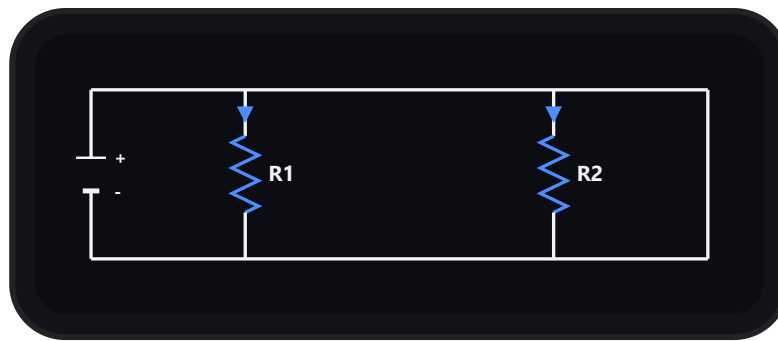


Diagram blueprint

circuit.parallel_resistors (electrical_circuit)

Visual semantic contract

visual-contract.parallel-circuit-branches

Canonical variant id

visual-contract.parallel-circuit-branches@1::branch_count=2,mode=both,show_branch_current_arrows=true,show_values=false::both

Linked assertion family

electrical.parallel_circuits

Linked assertion identifiers

EL-PARALLEL-RESISTANCE-CALC-001, EL-PARALLEL-RESISTANCE-001, EL-CIRCUIT-PARALLEL-STRUCTURE-001, EL-PARALLEL-CURRENT-CALC-001, EL-PARALLEL-DOMINANT-RESISTOR-001

Linked capability

cap.parallel.calculate_total_resistance, cap.parallel.solve_missing_branch, cap.parallel.recognise_structure, cap.parallel.calculate_branch_current, cap.parallel.identify_dominant_branch

Representation role

essential

MUST SHOW

- two shared horizontal rails
- every branch drawn as a separate vertical connection between the rails, labelled R1..Rn

MUST NOT SHOW

- branches drawn as a single series loop
- numeric component values

Show resistors connected as independent branches between two shared rails.



Diagram blueprint

circuit.parallel_resistors (electrical_circuit)

Visual semantic contract

visual-contract.parallel-circuit-branches

Canonical variant id

visual-contract.parallel-circuit-branches@1::branch_count=3,mode=both,show_branch_current_arrows=true,show_values=false::both

Linked assertion family

electrical.parallel_circuits

Linked assertion identifiers

EL-PARALLEL-RESISTANCE-CALC-001, EL-PARALLEL-RESISTANCE-001, EL-CIRCUIT-PARALLEL-STRUCTURE-001, EL-PARALLEL-CURRENT-CALC-001, EL-PARALLEL-DOMINANT-RESISTOR-001

Linked capability

cap.parallel.calculate_total_resistance, cap.parallel.solve_missing_branch, cap.parallel.recognise_structure, cap.parallel.calculate_branch_current, cap.parallel.identify_dominant_branch

Representation role

essential

MUST SHOW

- two shared horizontal rails
- every branch drawn as a separate vertical connection between the rails, labelled R1..Rn

MUST NOT SHOW

- branches drawn as a single series loop
- numeric component values

Show resistors connected as independent branches between two shared rails.

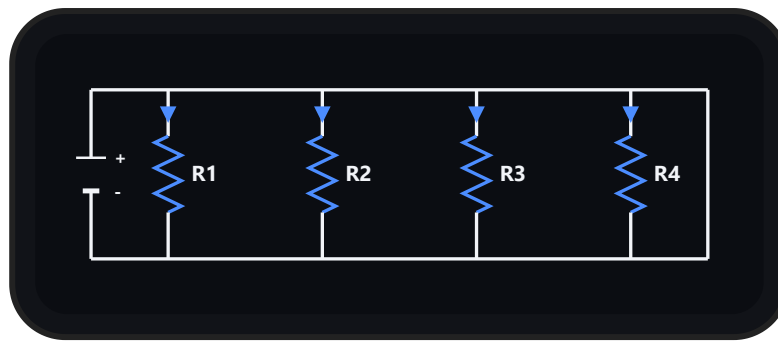


Diagram blueprint

circuit.parallel_resistors (electrical_circuit)

Visual semantic contract

visual-contract.parallel-circuit-branches

Canonical variant id

visual-contract.parallel-circuit-branches@1::branch_count=4,mode=both,show_branch_current_arrows=true,show_values=false::both

Linked assertion family

electrical.parallel_circuits

Linked assertion identifiers

EL-PARALLEL-RESISTANCE-CALC-001, EL-PARALLEL-RESISTANCE-001, EL-CIRCUIT-PARALLEL-STRUCTURE-001, EL-PARALLEL-CURRENT-CALC-001, EL-PARALLEL-DOMINANT-RESISTOR-001

Linked capability

cap.parallel.calculate_total_resistance, cap.parallel.solve_missing_branch, cap.parallel.recognise_structure, cap.parallel.calculate_branch_current, cap.parallel.identify_dominant_branch

Representation role

essential

MUST SHOW

- two shared horizontal rails
- every branch drawn as a separate vertical connection between the rails, labelled R1..Rn

MUST NOT SHOW

- branches drawn as a single series loop
- numeric component values

Contrast a fixed pulley (direction change only, $MA=1$) with a simple movable pulley (MA approximately 2, via two supporting rope segments) so the learner recognises the effort force/distance trade-off.

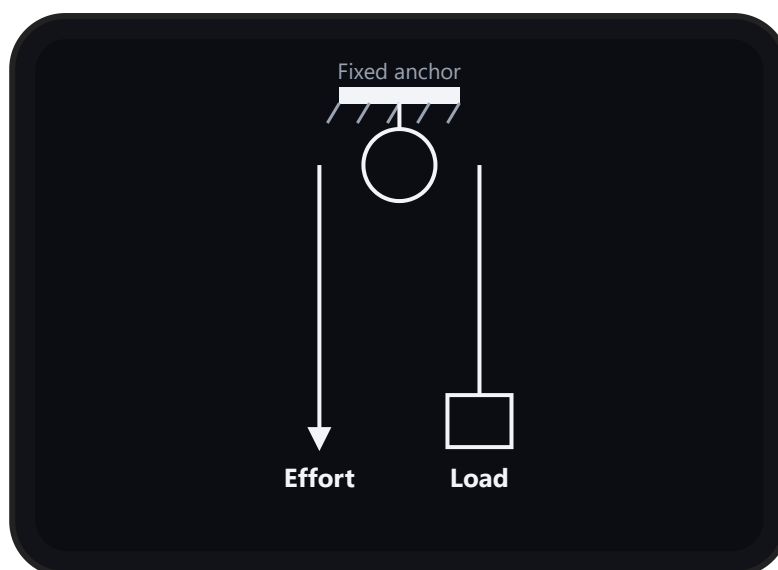


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

mechanical.pulley_arrangement (mechanical)

visual-contract.pulley-fixed-vs-movable

visual-contract.pulley-fixed-vs-movable@1::arrangement=fixed,mode=both::both

foundational.levers_mechanical_advantage

FP-CONCEPT-PULLEY-001, FP-PULLEY-FIXED-VS-MOVABLE-001, FP-REL-PULLEY-MECHANICAL-ADVANTAGE-001, FP-REL-PULLEY-FORCE-DISTANCE-TRADEOFF-001

cap.foundational.pulleys.recognise

essential

MUST SHOW

- the pulley wheel
- a fixed anchor point, labelled
- Effort and Load, both labelled
- for 'movable', two visually distinct rope segments both connecting to the movable pulley/load

MUST NOT SHOW

- a multi-pulley block-and-tackle system (out of governed scope -- only one fixed and one simple movable example)
- a physically-accurate rope-wrap path (a stylised schematic of straight segments is this course's house style)

Contrast a fixed pulley (direction change only, $MA=1$) with a simple movable pulley (MA approximately 2, via two supporting rope segments) so the learner recognises the effort force/distance trade-off.

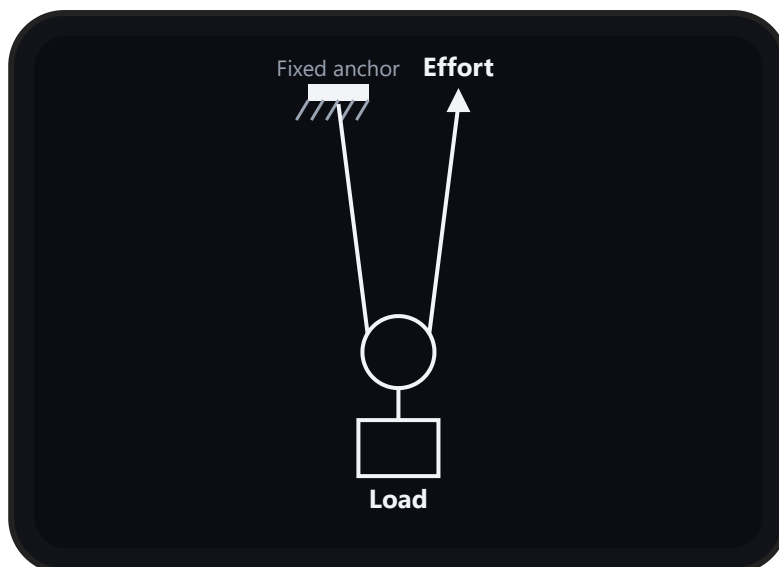


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

mechanical.pulley_arrangement (mechanical)

visual-contract.pulley-fixed-vs-movable

visual-contract.pulley-fixed-vs-movable@1::arrangement=movable,mode=both::both

foundational.levers_mechanical_advantage

FP-CONCEPT-PULLEY-001, FP-PULLEY-FIXED-VS-MOVABLE-001, FP-REL-PULLEY-MECHANICAL-ADVANTAGE-001, FP-REL-PULLEY-FORCE-DISTANCE-TRADEOFF-001

cap.foundational.pulleys.recognise

essential

MUST SHOW

- the pulley wheel
- a fixed anchor point, labelled
- Effort and Load, both labelled
- for 'movable', two visually distinct rope segments both connecting to the movable pulley/load

MUST NOT SHOW

- a multi-pulley block-and-tackle system (out of governed scope -- only one fixed and one simple movable example)
- a physically-accurate rope-wrap path (a stylised schematic of straight segments is this course's house style)

Show two conductor rods differing only in length (or only in cross-sectional area) so the learner predicts the qualitative effect on resistance, without any numeric $R = \rho L / A$ calculation.

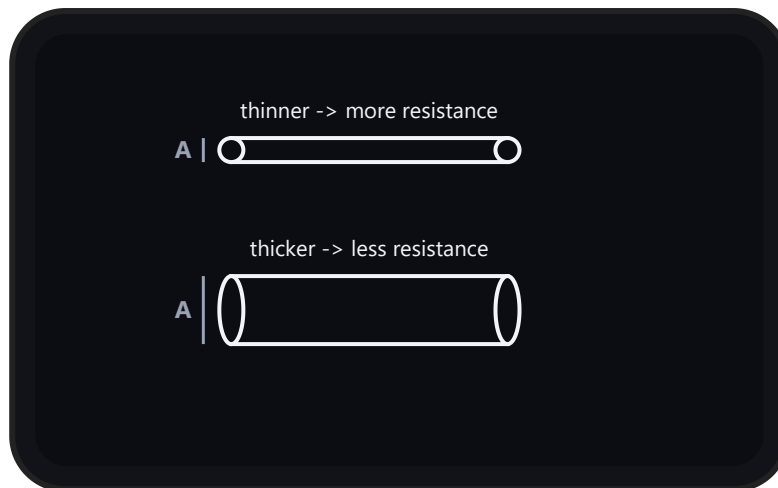


Diagram blueprint

mechanical.resistivity_dimensions (mechanical)

Visual semantic contract

visual-contract.resistivity-length-area-dimensions

Canonical variant id

visual-contract.resistivity-length-area-dimensions@1::comparison=area,mode=both::both

Linked assertion family

electrical.resistivity

Linked assertion identifiers

EL-RESISTIVITY-LENGTH-EFFECT-001, EL-RESISTIVITY-AREA-EFFECT-001, EL-CONDUCTOR-RESISTANCE-FACTORS-001

Linked capability

cap.resistivity.predict_length_effect, cap.resistivity.predict_area_effect

Representation role

essential

MUST SHOW

- two rod/cylinder schematics side by side
- a symbolic caption on each rod stating the qualitative resistance consequence

MUST NOT SHOW

- any numeric length, area or resistance value (symbolic captions only, per CC-05A valueEmbedding policy)
- an actual $R = \rho L / A$ calculation (out of this diagram's governed scope -- handled by formula.resistivity elsewhere)

Show two conductor rods differing only in length (or only in cross-sectional area) so the learner predicts the qualitative effect on resistance, without any numeric $R = \rho L / A$ calculation.

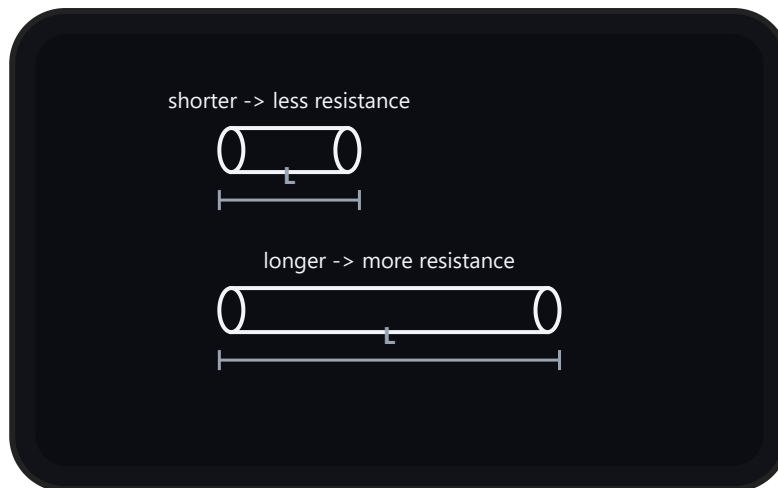


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

mechanical.resistivity_dimensions (mechanical)

visual-contract.resistivity-length-area-dimensions

visual-contract.resistivity-length-area-dimensions@1::comparison=length,mode=both::both

electrical.resistivity

EL-RESISTIVITY-LENGTH-EFFECT-001, EL-RESISTIVITY-AREA-EFFECT-001, EL-CONDUCTOR-RESISTANCE-FACTORS-001

cap.resistivity.predict_length_effect, cap.resistivity.predict_area_effect

essential

MUST SHOW

- two rod/cylinder schematics side by side
- a symbolic caption on each rod stating the qualitative resistance consequence

MUST NOT SHOW

- any numeric length, area or resistance value (symbolic captions only, per CC-05A valueEmbedding policy)
- an actual $R = \rho L / A$ calculation (out of this diagram's governed scope -- handled by formula.resistivity elsewhere)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

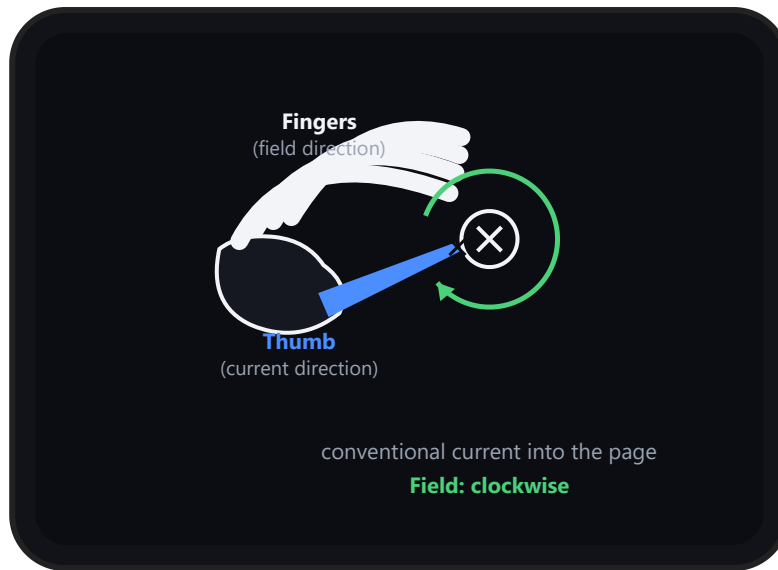


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=into_page,field_rotation=clockwise,mode=teaching,show_field_arrows=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

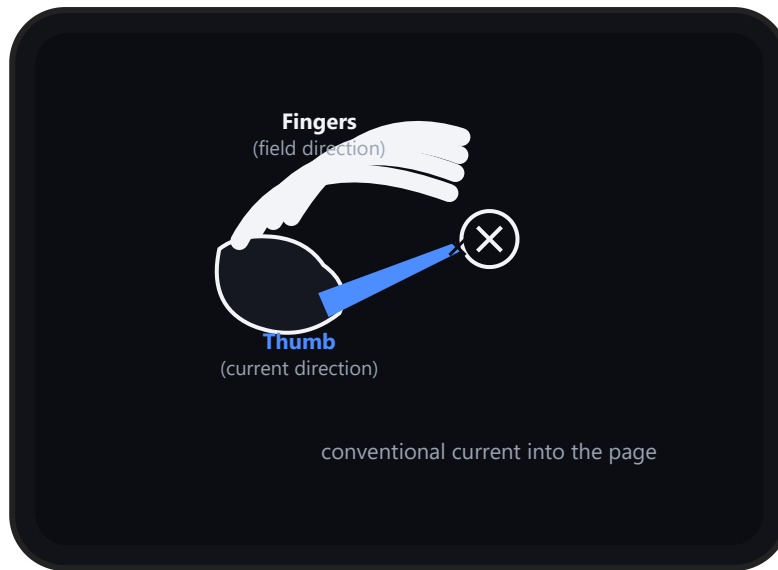


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=into_page,mode=assessment,show_field_arrows=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

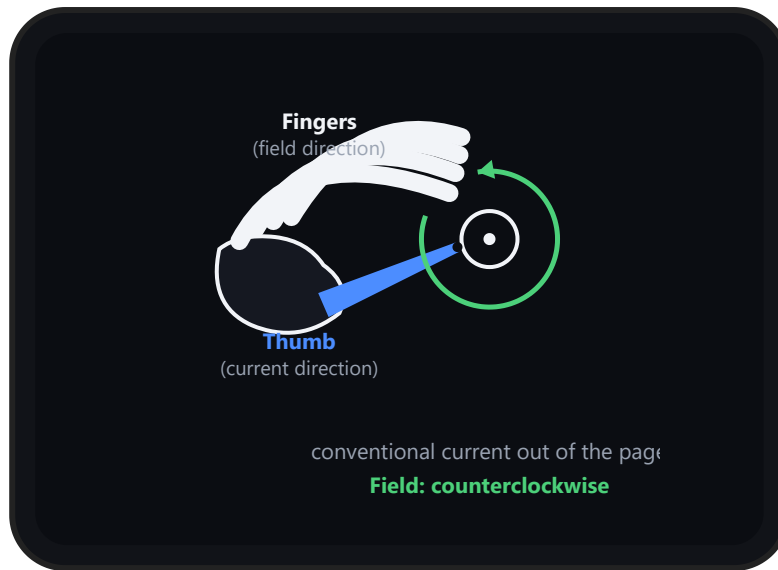


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=out_of_page,field_rotation=counterclockwise,mode=teaching,show_field_arrows=true::teaching

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Teach the right-hand grip rule for the magnetic field circulating around a current-carrying conductor.

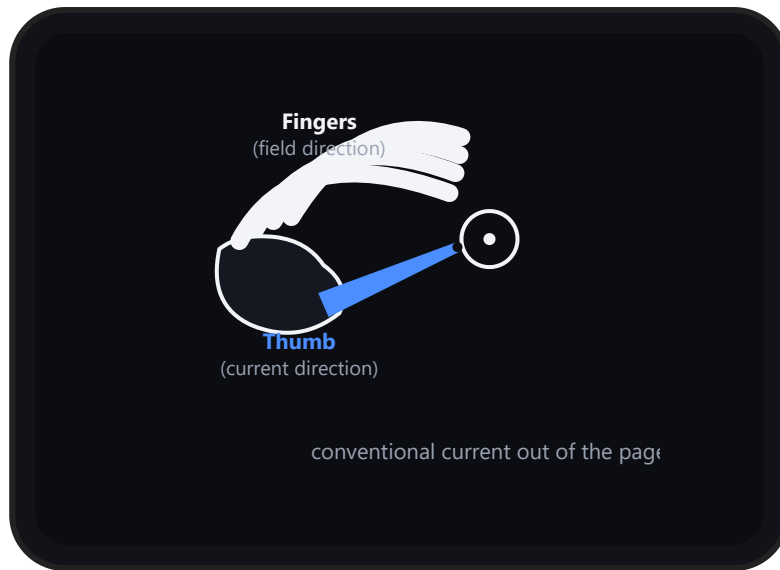


Diagram blueprint

Visual semantic contract

Canonical variant id

Linked assertion family

Linked assertion identifiers

Linked capability

Representation role

magnetic.field_conductor_direction (magnetic_field)

visual-contract.right-hand-grip-rule

visual-contract.right-hand-grip-rule@1::current_direction=out_of_page,mode=assessment,show_field_arrows=true::assessment

electrical.magnetism_and_electromagnetism

EL-CONCEPT-MAGNETIC-FIELD-CURRENT-001

cap.magnetism.interpret_field_direction

essential

MUST SHOW

- a recognisable right hand
- a distinctly labelled thumb
- distinctly labelled curled fingers
- the conductor cross-section the hand grips

MUST NOT SHOW

- a generic force arrow presented as if it were the field direction (the pre-CC-05C-DIAGRAM-FIX defect: a generic magnetic-force diagram shown for a right-hand-grip-rule lesson)
- the field-rotation arrow/label in assessment mode (it is the assessed answer)

Show resistors connected one after another in a single loop, with an arrow indicating the single shared current direction around the loop.



Diagram blueprint

circuit.series_resistors (electrical_circuit)

Visual semantic contract

visual-contract.series-circuit-current-direction

Canonical variant id

visual-contract.series-circuit-current-direction@1::component_count=2,mode=both,show_current_arrow=true,show_values=false::both

Linked assertion family

electrical.series_circuits

Linked assertion identifiers

EL-SERIES-RESISTANCE-CALC-001, EL-SERIES-RESISTANCE-001, EL-SERIES-VOLTAGE-CALC-001, EL-SERIES-DOMINANT-RESISTOR-001, EL-CIRCUIT-SERIES-STRUCTURE-001

Linked capability

cap.series.calculate_total_resistance, cap.series.solve_missing_component, cap.series.calculate_voltage_drop, cap.series.identify_dominant_component, cap.series.recognise_structure

Representation role

essential

MUST SHOW

- one continuous single loop
- every resistor drawn in the loop, labelled R1..Rn in order
- a current-direction arrow on the loop, pointing along the wire it sits on

MUST NOT SHOW

- a second, parallel current path
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show resistors connected one after another in a single loop, with an arrow indicating the single shared current direction around the loop.

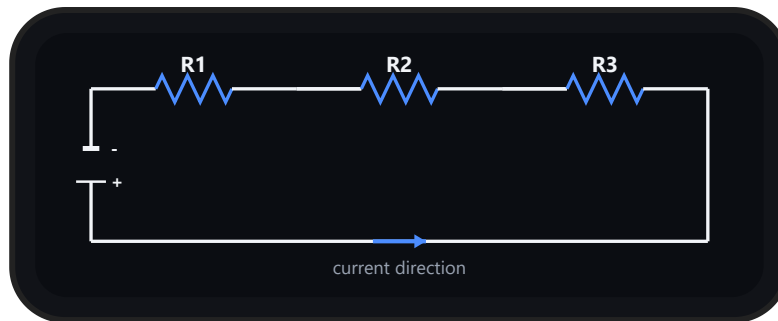


Diagram blueprint

`circuit.series_resistors (electrical_circuit)`

Visual semantic contract

`visual-contract.series-circuit-current-direction`

Canonical variant id

`visual-contract.series-circuit-current-direction@1::component_count=3,mode=both,show_current_arrow=true,show_values=false::both`

Linked assertion family

`electrical.series_circuits`

Linked assertion identifiers

`EL-SERIES-RESISTANCE-CALC-001, EL-SERIES-RESISTANCE-001, EL-SERIES-VOLTAGE-CALC-001, EL-SERIES-DOMINANT-RESISTOR-001, EL-CIRCUIT-SERIES-STRUCTURE-001`

Linked capability

`cap.series.calculate_total_resistance, cap.series.solve_missing_component, cap.series.calculate_voltage_drop, cap.series.identify_dominant_component, cap.series.recognise_structure`

Representation role

`essential`**MUST SHOW**

- one continuous single loop
- every resistor drawn in the loop, labelled R1..Rn in order
- a current-direction arrow on the loop, pointing along the wire it sits on

MUST NOT SHOW

- a second, parallel current path
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show resistors connected one after another in a single loop, with an arrow indicating the single shared current direction around the loop.

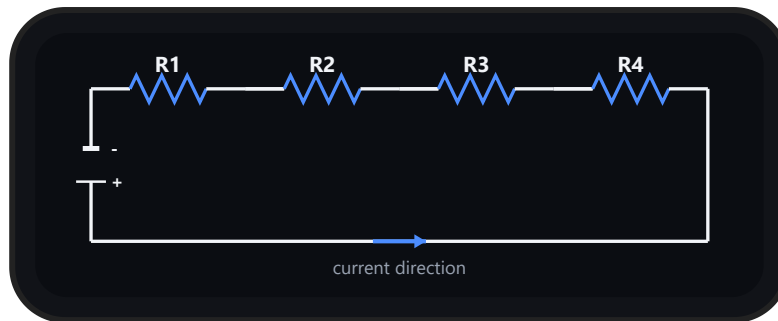


Diagram blueprint

circuit.series_resistors (electrical_circuit)

Visual semantic contract

visual-contract.series-circuit-current-direction

Canonical variant id

visual-contract.series-circuit-current-direction@1::component_count=4,mode=both,show_current_arrow=true,show_values=false::both

Linked assertion family

electrical.series_circuits

Linked assertion identifiers

EL-SERIES-RESISTANCE-CALC-001, EL-SERIES-RESISTANCE-001, EL-SERIES-VOLTAGE-CALC-001, EL-SERIES-DOMINANT-RESISTOR-001, EL-CIRCUIT-SERIES-STRUCTURE-001

Linked capability

cap.series.calculate_total_resistance, cap.series.solve_missing_component, cap.series.calculate_voltage_drop, cap.series.identify_dominant_component, cap.series.recognise_structure

Representation role

essential

MUST SHOW

- one continuous single loop
- every resistor drawn in the loop, labelled R1..Rn in order
- a current-direction arrow on the loop, pointing along the wire it sits on

MUST NOT SHOW

- a second, parallel current path
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show a genuinely mixed series-and-parallel topology so learners can distinguish it from pure series or pure parallel, and trace the current path(s) through it.

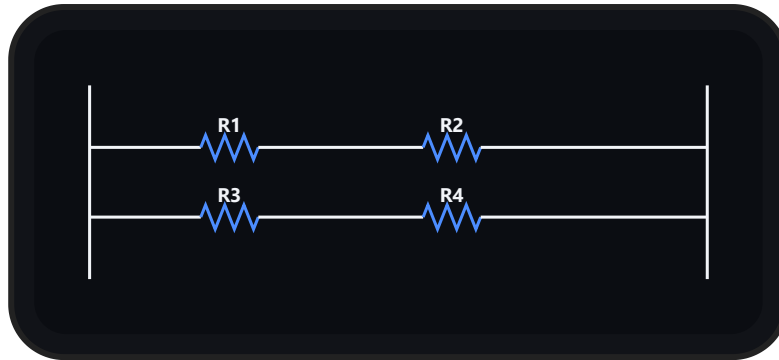


Diagram blueprint

circuit.series_parallel_mixed (electrical_circuit)

Visual semantic contract

visual-contract.series-parallel-mixed-topology

Canonical variant id

visual-contract.series-parallel-mixed-topology@1::branch_arrangement=parallel_of_series,mode=both,show_values=false::both

Linked assertion family

electrical.series_vs_parallel_comparison

Linked assertion identifiers

EL-CIRCUIT-SELECT-CONFIGURATION-001, EL-CIRCUIT-MIXED-SERIES-PARALLEL-RECOGNITION-001, EL-CIRCUIT-TRACE-CURRENT-PATH-001

Linked capability

cap.comparison.identify_topology, cap.comparison.recognise_mixed_circuit, cap.comparison.trace_current_path

Representation role

essential

MUST SHOW

- a genuine series run (one or more components in a single current path)
- a genuine parallel group (two or more components between shared local rails) within the same image
- symbolic R1..Rn labels in a single consistent reading order
- the specific arrangement (series-of-parallel vs parallel-of-series) visually distinguishable from the other

MUST NOT SHOW

- a topology that is actually pure series or pure parallel dressed up to look mixed
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)

Show a genuinely mixed series-and-parallel topology so learners can distinguish it from pure series or pure parallel, and trace the current path(s) through it.

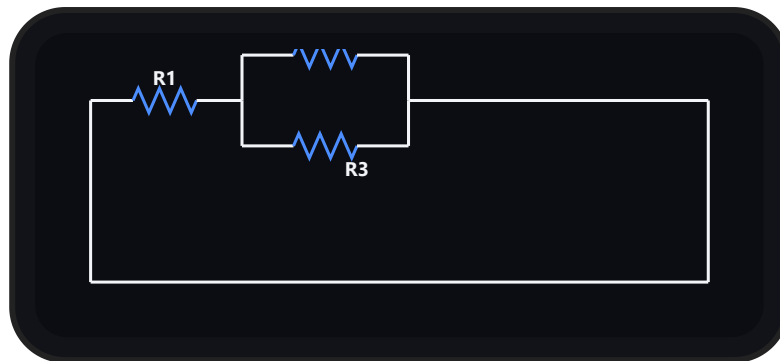


Diagram blueprint

circuit.series_parallel_mixed (electrical_circuit)

Visual semantic contract

visual-contract.series-parallel-mixed-topology

Canonical variant id

visual-contract.series-parallel-mixed-topology@1::branch_arrangement=series_of_parallel,mode=both,show_values=false::both

Linked assertion family

electrical.series_vs_parallel_comparison

Linked assertion identifiers

EL-CIRCUIT-SELECT-CONFIGURATION-001, EL-CIRCUIT-MIXED-SERIES-PARALLEL-RECOGNITION-001, EL-CIRCUIT-TRACE-CURRENT-PATH-001

Linked capability

cap.comparison.identify_topology, cap.comparison.recognise_mixed_circuit, cap.comparison.trace_current_path

Representation role

essential

MUST SHOW

- a genuine series run (one or more components in a single current path)
- a genuine parallel group (two or more components between shared local rails) within the same image
- symbolic R1..Rn labels in a single consistent reading order
- the specific arrangement (series-of-parallel vs parallel-of-series) visually distinguishable from the other

MUST NOT SHOW

- a topology that is actually pure series or pure parallel dressed up to look mixed
- numeric component values (symbolic labels only, per CC-05A valueEmbedding policy)